

YUNUS EMRE BALCI

Horsens / Odense, Denmark

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PROFILE

Full-stack engineer and applied ML researcher who ships fast and owns features end-to-end. Architected a self-hostable Discord alternative in Go with Redis pub/sub and ScyllaDB; sole developer of an 11,000+ line cross-platform mobile app; designed an RL architecture that trained 28.8× faster than CLIP-based baselines. Comfortable across Go and Python backends, React/TypeScript frontends, and containerized cloud deployments. Trilingual (Danish, Turkish, English) with a proven async-first remote workflow.

TECHNICAL SKILLS

Languages: Go, Python, JavaScript/TypeScript, C#, C++, Java, SQL, Bash, Lua

Backend & Data: Go (Fiber), Node.js, Redis, PostgreSQL, ScyllaDB, MongoDB, GraphQL, REST, WebSocket, gRPC

Cloud & DevOps: Docker, Kubernetes, GCP, AWS, CI/CD, Linux (Debian/Ubuntu), Vercel, Render

Frontend & Mobile: React, Next.js, React Native, TypeScript, Three.js, Tailwind CSS, Tauri

AI / ML: PyTorch, Reinforcement Learning (PPO, SAC), MuJoCo, HuggingFace Transformers, Groq API, CUDA

Game Dev: Unity 6 (URP), C#, GameMaker Studio 2

EXPERIENCE

Software Engineer, AI Training (Contractor)

Apr 2026 – Present

micro1 – Remote (San Francisco, USA)

- Contracting with micro1, a San Francisco-based AI talent platform, on an enterprise AI agent initiative
- Reviewing and authoring software engineering prompts and reference implementations used to train and evaluate AI coding agents
- Evaluating model outputs against engineering best practices; providing structured feedback on correctness, style, and reasoning across multi-step tasks
- Operating asynchronously across time zones in a distributed team of senior technical experts

Freelance Software Developer

2023 – Present

Self-employed – Remote, Denmark

- Designing and shipping production websites for Danish SMBs; full ownership of discovery, build, deployment, and maintenance
- Recent client: *Mortens Chokolade* (Odense) – rebuilt the storefront on Shopify (Dawn theme) with a Von Linné-inspired editorial design system across 15+ pages, including custom Liquid templates and product page redesign
- Stack: React, Next.js, Node.js, MongoDB, Shopify Liquid; SEO, performance, and async client communication

Technical Sales Consultant

Jun 2022 – Apr 2025

Telia / Norlys – Denmark

- Technical consulting on mobile, broadband, and network solutions across omnichannel touchpoints
- Balanced a client-facing role while completing computer science studies

PROJECTS

Nexus – Real-Time Communication Platform

Go (Fiber), Redis, ScyllaDB, WebSocket, Docker – 2025

- Architected and built a self-hostable Discord alternative with a Go backend, Redis pub/sub for cross-instance messaging, and ScyllaDB for persistent storage
- Engineered WebSocket hub with channel-based broadcasting, presence tracking, and JWT authentication middleware
- Implemented voice chat infrastructure with a custom SFU (Selective Forwarding Unit) for WebRTC signaling
- Built React + Tauri desktop client with real-time state sync, theming engine, and plugin architecture

- Containerized full stack with Docker; designed for horizontal scaling across multiple server instances via Redis pub/sub

MotionRL – Motion-Language Control System

Python, PyTorch, MuJoCo – 2025

Research supervised by Prof. Melih Kandemir, SDU

- Designed a hierarchical RL architecture enabling humanoid agents to execute motions from natural-language commands
- Achieved **28.8× faster** reward computation and **32.5× lower** GPU memory vs. CLIP-based approaches
- Demonstrated 97–99% zero-shot generalization to unseen motion descriptions

Contraband Chef – First-Person Cooking Crime Simulator

Unity 6 (URP), C# – In Development

- Building a singleplayer game where players cook, serve, and expand a criminal food empire in an open-world city
- Engineered physics-based food handling system with station interactions, assembly plates, and order management
- Implemented FPS controller, dynamic HUD, serving window logic, and camera-anchor zoom for cooking stations

Narrator AI – Real-Time Twitch Streaming Companion

Python, Groq API, Three.js, WebSocket – 2025

- Built an AI co-commentator for live Twitch streams using vision models, speech-to-text, and text-to-speech
- Engineered Three.js browser overlay served via aiohttp/WebSocket for animated on-stream presence

MyDrive – Cross-Platform Vehicle Management App

React Native, Firebase, REST APIs – 2024–2025

- Sole developer of a mobile app for the Danish vehicle market – 11,000+ lines across 20+ screens
- Integrated Danish vehicle registry APIs, OCR receipt scanning, and real-time fuel price aggregation
- Architected serverless backend with Firebase Auth, Firestore, and Cloud Functions

Unearth – Multiplayer 3D Browser Game

Three.js, Node.js, Socket.io – 2024

- Shipped a GeoGuessr-style archaeology game with procedural terrain, 3D museum artifacts, and competitive multiplayer
- Built real-time lobby system, animation pipeline, and risk-reward scoring engine

CTF Security Infrastructure

Python, SQL, Docker – Bachelor Thesis, Grade: 10

- Developed Capture the Flag challenges spanning ML exploitation, SQL injection, and Linux privilege escalation
- Built dynamic scoring backend with fully containerized challenge deployment

EDUCATION

BSc Computer Science

2022 – Expected 2026

University of Southern Denmark (SDU)

152.5 of 180 ECTS completed. Bachelor Thesis: *Investigating and Creating CTF Challenges* – Grade: 10.

Graduate Coursework in Computer Science

2025

University of Southern Denmark (SDU)

Supervised research project (10 ECTS) in reinforcement learning and motion-language control under Prof. Melih Kandemir.

ADDITIONAL

Languages: Danish (native), Turkish (native), English (fluent, C2)

Availability: Immediately available for fully remote roles; open to periodic travel

Portfolio: 20+ active repositories at github.com/YunusBalci21